

WEEKLY REPORT and MEETING AGENDA

Report #: 1
Date: 2/23/2026

Project Name: Emergency Response Mapping Sphere
Prepared by: Hai Yen Nguyen

Agenda for the meeting

1. Discuss objectives and deliverables for the next 2 upcoming weeks
2. Discuss any project concerns
3. Review hardware compatibility (Raspberry Pi, camera multiplexer, IR module)
4. Power system design discussion
5. Define responsibilities and integration timeline

Overall accomplishments since last meeting

1. Task Delegation
2. Component Ordering (Raspberry Pi, Camera Module, Camera Multiplexer)
3. Discussion on IR camera implementation & alternatives to address smoke
4. Created initial system architecture diagram
5. Conducted research on power requirements for multi-camera + Pi setup

Tasks completed by each team member since last meeting

Task description	Assigned to	Completed?
Researched camera multiplexer compatibility	Josh	yes
Compared IR vs. thermal camera modules	Kiet	yes
Ordered hardware components	Brandon	yes
Designed System architecture	Antony	yes
Evaluated power bank options for stability, capacity, and fit	Hai Yen	yes

Plans for next period

1. Order a compatible power bank for the sphere (5V 3A minimum output).
2. Meeting with Jack to discuss project implementation strategy.
3. Begin hardware assembly once components arrive.
4. Test camera switching functionality using multiplexer.

Task assignment per team member (to be completed before the next meeting)

Task description	Assigned to
Verify power bank voltage/current stability	Antony
Write initial camera interface test script	Josh
Integrate multiplexer with Raspberry Pi	Kiet
Investigate enclosure thermal management	Brandon
Draft risk assessment document	Hai Yen

Project management status

1. Hardware ordered and expected to come in within the next week
2. Roles are defined and task delegation is clear
3. Purchased Raspberry Pi 4B, camera module, and camera multiplexer
- 4.
- 5.

Minutes from previous meeting

Please be thorough, a bullet list of 2-3 single-liners is not sufficient

During the previous meeting, the team focused on clarifying the system architecture for the spherical camera device. We discussed the feasibility of using multiple camera modules connected through a multiplexer to a single Raspberry Pi 4B. The team evaluated whether the Pi would have sufficient processing power and bandwidth to handle camera switching efficiently.

We also discussed the smoke interference problem and explored potential solutions. One proposed solution was incorporating IR cameras to improve visibility in low-visibility environments. However, concerns were raised regarding the effectiveness of IR illumination in penetrating dense smoke, prompting further evaluation of its feasibility within our design constraints

Additionally, the power system was identified as a critical design constraint. Since the device must operate inside a fully enclosed sphere without a direct connection to an external power source, the team agreed that a compact, high-output power bank would be necessary. We discussed the required voltage (5V) and minimum current output (3A) to ensure stable operation when multiple components are active.

Finally, tasks were delegated to ensure progress before the next meeting, including component ordering, compatibility research, and drafting a preliminary hardware diagram.